

Cálculo de $q_\pi(s, a) = \mathbb{E}[G_t \mid S_t = s, A_t = a]$ por marginalización del modelo gráfico del MDP

1 Convención de Sutton & Barto

Variables aleatorias en orden temporal:

$$S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}, R_{t+2}, S_{t+2}, A_{t+2}, R_{t+3}, \dots$$

con dependencias

$$A_{t+j} \sim \pi(\cdot \mid S_{t+j}), \quad S_{t+j+1} \sim p(\cdot \mid S_{t+j}, A_{t+j}), \quad R_{t+j+1} \sim p(\cdot \mid S_{t+j}, A_{t+j}, S_{t+j+1}).$$

El retorno es

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

En $q_\pi(s, a)$ condicionamos en $S_t = s$ y $A_t = a$. La diferencia con $v_\pi(s)$ es que A_t ya no se marginaliza con $\pi(\cdot \mid s)$: su valor está fijado en a . Las acciones posteriores $A_{t+1}, A_{t+2}, \dots$ sí se marginalizan con π .

2 La esperanza como suma contra la conjunta

$$\mathbb{E}[G_t \mid S_t = s, A_t = a] = \sum_{r_{t+1}, r_{t+2}, \dots} \left(\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \right) p(R_{t+1} = r_{t+1}, R_{t+2} = r_{t+2}, \dots \mid S_t = s, A_t = a).$$

3 Reagrupar por índice de recompensa

$$\begin{aligned} \mathbb{E}[G_t \mid S_t = s, A_t = a] &= \sum_{k=0}^{\infty} \gamma^k \sum_{r_{t+1}, r_{t+2}, \dots} r_{t+k+1} \\ &\quad \times p(R_{t+1} = r_{t+1}, R_{t+2} = r_{t+2}, \dots \mid S_t = s, A_t = a). \end{aligned}$$

Para cada k fijo, sumar la conjunta sobre todas las recompensas excepto R_{t+k+1} da la marginal $p(R_{t+k+1} = r \mid S_t = s, A_t = a)$:

$$\mathbb{E}[G_t \mid S_t = s, A_t = a] = \sum_{k=0}^{\infty} \gamma^k \sum_r r p(R_{t+k+1} = r \mid S_t = s, A_t = a).$$

4 Marginalizar cada recompensa individualmente

Fijamos k . La recompensa R_{t+k+1} depende de $(S_{t+k}, A_{t+k}, S_{t+k+1})$. La conjunta sobre las variables relevantes, dado $S_t = s$ y $A_t = a$, es

$$\begin{aligned} &p(S_{t+1}, R_{t+1}, A_{t+1}, \dots, S_{t+k}, A_{t+k}, S_{t+k+1}, R_{t+k+1} \mid S_t = s, A_t = a) \\ &= p(S_{t+1} \mid s, a) p(R_{t+1} \mid s, a, S_{t+1}) \dots \\ &\quad \times \pi(A_{t+k} \mid S_{t+k}) p(S_{t+k+1} \mid S_{t+k}, A_{t+k}) p(R_{t+k+1} \mid S_{t+k}, A_{t+k}, S_{t+k+1}). \end{aligned}$$

Nótese que $A_t = a$ está fijado, así que no aparece el factor $\pi(A_t \mid s)$; la primera transición usa directamente $p(S_{t+1} \mid s, a)$.

Sumamos sobre todas las variables excepto R_{t+k+1} . Ocurren dos cosas:

1. **Las recompensas intermedias se eliminan.** Para cada $j = 0, \dots, k-1$,

$$\sum_{r_{t+j+1}} p(R_{t+j+1} = r_{t+j+1} \mid S_{t+j} = s_{t+j}, A_{t+j} = a_{t+j}, S_{t+j+1} = s_{t+j+1}) = 1.$$

2. **Los estados y las acciones futuras permanecen.** Queda la cadena

$$p(R_{t+k+1} = r \mid S_t = s, A_t = a) = \sum_{s_{t+1}, a_{t+1}, \dots, s_{t+k}, a_{t+k}, s_{t+k+1}} p(s_{t+1} \mid s, a) \\ \times \left[\prod_{j=1}^{k-1} \pi(a_{t+j} \mid s_{t+j}) p(s_{t+j+1} \mid s_{t+j}, a_{t+j}) \right] \pi(a_{t+k} \mid s_{t+k}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}) p(R_{t+k+1} = r \mid s_{t+k}, a_{t+k}, s_{t+k+1}).$$

5 Sustituir en la esperanza

$$\mathbb{E}[R_{t+k+1} \mid S_t = s, A_t = a] = \sum_{s_{t+1}, a_{t+1}, \dots, s_{t+k}, a_{t+k}, s_{t+k+1}} p(s_{t+1} \mid s, a) \\ \times \left[\prod_{j=1}^{k-1} \pi(a_{t+j} \mid s_{t+j}) p(s_{t+j+1} \mid s_{t+j}, a_{t+j}) \right] \pi(a_{t+k} \mid s_{t+k}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}) \bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}),$$

donde

$$\bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}) = \sum_r r p(R_{t+k+1} = r \mid s_{t+k}, a_{t+k}, s_{t+k+1}).$$

6 Primeros términos explícitos

$k = 0$ (recompensa R_{t+1}). R_{t+1} depende de (S_t, A_t, S_{t+1}) , con $S_t = s$ y $A_t = a$ fijos; sólo se marginaliza S_{t+1} :

$$\mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] = \sum_{s_{t+1}} p(s_{t+1} \mid s, a) \bar{R}(s, a, s_{t+1}).$$

$k = 1$ (recompensa R_{t+2}). R_{t+2} depende de $(S_{t+1}, A_{t+1}, S_{t+2})$; se marginaliza $S_{t+1}, A_{t+1}, S_{t+2}$:

$$\mathbb{E}[R_{t+2} \mid S_t = s, A_t = a] = \sum_{s_{t+1}, a_{t+1}, s_{t+2}} p(s_{t+1} \mid s, a) \\ \times \pi(a_{t+1} \mid s_{t+1}) p(s_{t+2} \mid s_{t+1}, a_{t+1}) \bar{R}(s_{t+1}, a_{t+1}, s_{t+2}).$$

$k = 2$ (recompensa R_{t+3}).

$$\mathbb{E}[R_{t+3} \mid S_t = s, A_t = a] = \sum_{s_{t+1}, a_{t+1}, s_{t+2}, a_{t+2}, s_{t+3}} p(s_{t+1} \mid s, a) \\ \times \pi(a_{t+1} \mid s_{t+1}) p(s_{t+2} \mid s_{t+1}, a_{t+1}) \pi(a_{t+2} \mid s_{t+2}) p(s_{t+3} \mid s_{t+2}, a_{t+2}) \bar{R}(s_{t+2}, a_{t+2}, s_{t+3}).$$

7 Ensamblar el retorno

$$q_\pi(s, a) = \mathbb{E}[G_t \mid S_t = s, A_t = a] = \sum_{k=0}^{\infty} \gamma^k \mathbb{E}[R_{t+k+1} \mid S_t = s, A_t = a],$$

es decir,

$$q_\pi(s, a) = \sum_{k=0}^{\infty} \gamma^k \sum_{s_{t+1}, a_{t+1}, \dots, s_{t+k}, a_{t+k}, s_{t+k+1}} p(s_{t+1} \mid s, a) \\ \times \left[\prod_{j=1}^{k-1} \pi(a_{t+j} \mid s_{t+j}) p(s_{t+j+1} \mid s_{t+j}, a_{t+j}) \right] \pi(a_{t+k} \mid s_{t+k}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}) \bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}).$$

8 Recuperar la ecuación de Bellman para q_π

Separamos $k = 0$:

$$q_\pi(s, a) = \underbrace{\sum_{s_{t+1}} p(s_{t+1} \mid s, a) \bar{R}(s, a, s_{t+1})}_{k=0} \\ + \sum_{k \geq 1} \gamma^k \sum_{s_{t+1}, a_{t+1}, \dots, s_{t+k+1}} p(s_{t+1} \mid s, a) \left[\prod_{j=1}^{k-1} \pi(a_{t+j} \mid s_{t+j}) p(s_{t+j+1} \mid s_{t+j}, a_{t+j}) \right] \\ \times \pi(a_{t+k} \mid s_{t+k}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}) \bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}).$$

Para $k \geq 1$, separamos la primera transición $s \rightarrow s_{t+1}$ y la primera acción futura a_{t+1} . La suma interior sobre $a_{t+1}, s_{t+2}, \dots, s_{t+k+1}$ es exactamente $v_\pi(s_{t+1})$ (la misma expresión con k desplazado una unidad):

$$\sum_{k \geq 1} \gamma^k \sum_{s_{t+1}} p(s_{t+1} \mid s, a) v_\pi(s_{t+1}),$$

donde

$$v_\pi(s_{t+1}) = \sum_{a_{t+1}} \pi(a_{t+1} \mid s_{t+1}) \sum_{s_{t+2}} p(s_{t+2} \mid s_{t+1}, a_{t+1}) \left[\bar{R}(s_{t+1}, a_{t+1}, s_{t+2}) + \gamma v_\pi(s_{t+2}) \right].$$

Combinando,

$$q_\pi(s, a) = \sum_{s'} p(s' \mid s, a) \left[\bar{R}(s, a, s') + \gamma v_\pi(s') \right].$$

Usando $v_\pi(s') = \sum_{a'} \pi(a' \mid s') q_\pi(s', a')$ (que se obtiene marginalizando A_{t+1} en la definición de v_π), también se tiene la forma alternativa

$$q_\pi(s, a) = \sum_{s'} p(s' \mid s, a) \left[\bar{R}(s, a, s') + \gamma \sum_{a'} \pi(a' \mid s') q_\pi(s', a') \right].$$

Observaciones

- La única diferencia con $v_\pi(s)$ es que $A_t = a$ está fijado: no se suma sobre A_t con $\pi(\cdot \mid s)$, y el primer factor es $p(s_{t+1} \mid s, a)$ en lugar de $\sum_{a_t} \pi(a_t \mid s) p(s_{t+1} \mid s, a_t)$.
- Las acciones futuras $A_{t+1}, A_{t+2}, \dots$ sí se marginalizan con π , porque la política sigue actuando a partir de $t + 1$.
- Cada recompensa conserva sus tres argumentos: S_{t+k} , A_{t+k} y S_{t+k+1} .
- Las recompensas intermedias se eliminan por normalización al marginalizarlas.
- En ningún paso se invoca la linealidad de la esperanza como propiedad: sólo se reagrupan sumas contra la conjunta y se usa $\sum_x p(x \mid \cdot) = 1$.